

# Purujit Kantiya

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## Education

### B.A.S.c - Mechatronics and Robotics Engineering

Sep 2024 – May 2028

Grade: 4.0

Queen's University

- Awards: Deans List, Ontario Engineering Competition 1st Place, Canadian Engineering Competition 3rd Place
- Extracurriculars: Aerospace Design Team, Hyperloop Design Team, QHacks, Merlin Neurotech

## Professional Experience

### Verification Engineering Intern- Xona Space Systems

May 2026 – Aug 2027

- Supporting user equipment verification for compatibility with Xona's PULSAR signal by developing test scripts in Python to check hardware against ICD requirements.
- Automating testing plans to streamline hardware testing pipeline.
- Migrated requirements for hardware verification from Excel to Flow Engineering and integrated automatic updates using the Python API

### Construction Engineering Intern - Tatham Engineering

May 2025 – Aug 2025

- Supported construction site inspections to monitor progress, verify quality, and ensure adherence to design specifications.
- Reviewed contractor submittals, including construction schedules, shop drawings, and material specifications, to ensure compliance with project requirements.
- Collected and analyzed survey data to confirm accuracy of contractor work and support progress verification.
- Provided technical guidance and recommendations to contractors on construction methods, sequencing, and compliance with municipal standards.

## Extracurricular Experience

### Captain - Queen's Aerospace Design Team

May 2026 – Present

- Leading 100 students in the development of several aerospace projects ranging from research papers to physical drones.
- Responsible for maintaining team resources so that technical goals can be met by other executive members.

### Simulation Team Member - Queen's Aerospace Design Team

Sep 2025 – Present

- Contributing to the development of a multi-drone swarm simulation for presentation at the 2026 ICUAS conference, using ROS2, Gazebo, and a Linux-based workflow.
- Designing and implementing ROS2 nodes that command the drones to scan and extract data from the surrounding environment.
- Using Git for version control, collaborative development, and code review to ensure reliable and maintainable simulation software.

### Systems Integration Team Manager - Queen's Aerospace Design Team

May 2025 – April 2026

- Leading a team of 10 students in the design, integration, and testing of electrical systems for a fixed wing UAV.
- Plan and conduct systems tests using custom test stands, sensor packages and software.
- Oversee component selection, procurement, and cost tracking to stay within project budget and timeline.
- Coordinate with structures and aerodynamics teams to align system requirements and optimize aircraft performance.

## Projects

### Autonomous Mobile Robot

Jan 2026 – Apr 2026

- Developed a ROS2-based autonomous mobile rover integrating sensing, control, mapping, and navigation on a Raspberry Pi platform using modular packages for bring-up, robot description (URDF/TF2), and LiDAR interfacing (rplidar\_ros).
- Implemented closed-loop PI motor control via Arduino serial communication, using wheel encoder feedback to estimate odometry and maintain stable motion control.
- Integrated 2D LiDAR-based SLAM (SLAM Toolbox) with encoder odometry to generate real-time occupancy grid maps for simultaneous localization and environment mapping.
- Implemented full autonomous navigation using the ROS2 Nav2 stack (NavFn global planner, regulated pure pursuit local controller, behavior trees), achieving successful goal-based autonomous navigation in a mapped environment.

## Technical Skills

**Design:** Solidworks CAD, Onshape, LTSpice, Altium, QGroundControl, KiCad, Altera Quartus II, Arduino

**Programming:** Python, C++, VHDL, ROS2, Linux, Docker, Git

**Productivity:** Word, PowerPoint, Excel

**Languages:** Sindhi (Fluent), English (Fluent)